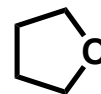
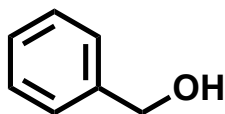
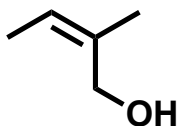
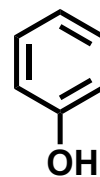
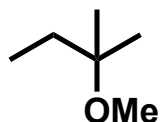
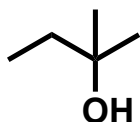
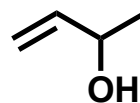
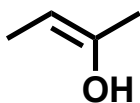
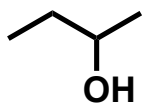


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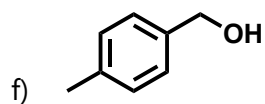
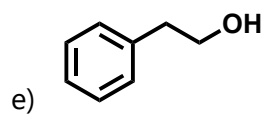
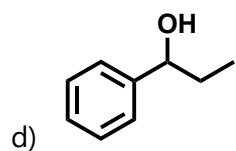
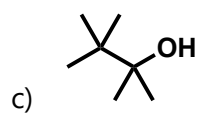
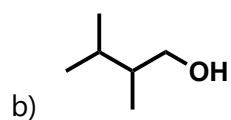
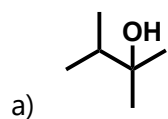
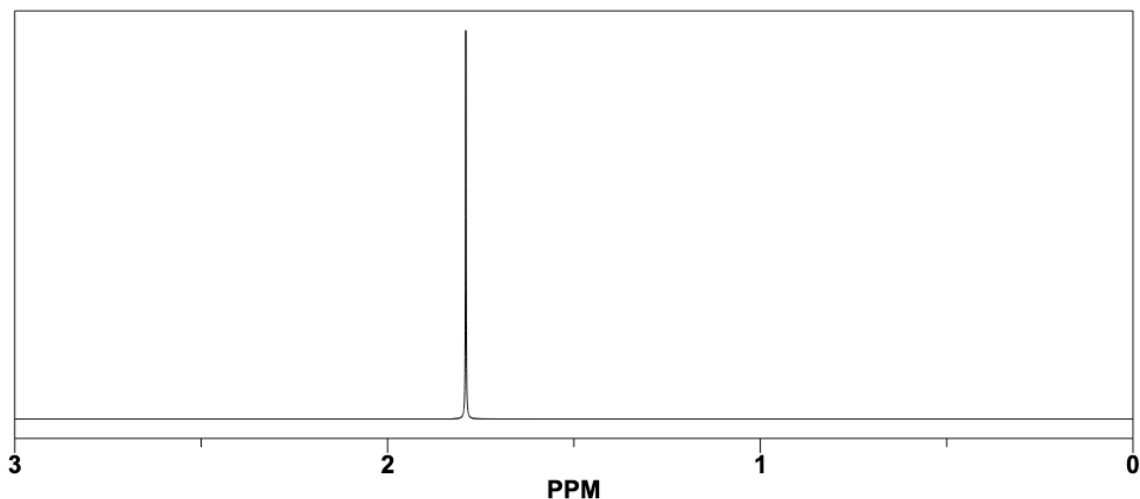
Activity 6: Dehydration of Cyclohexanol

Extra Practice Problems

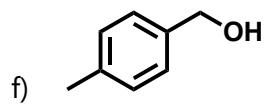
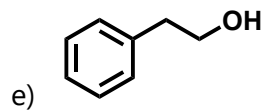
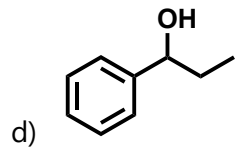
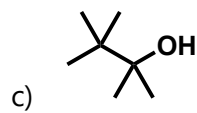
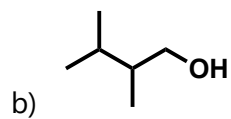
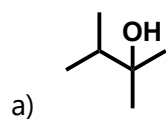
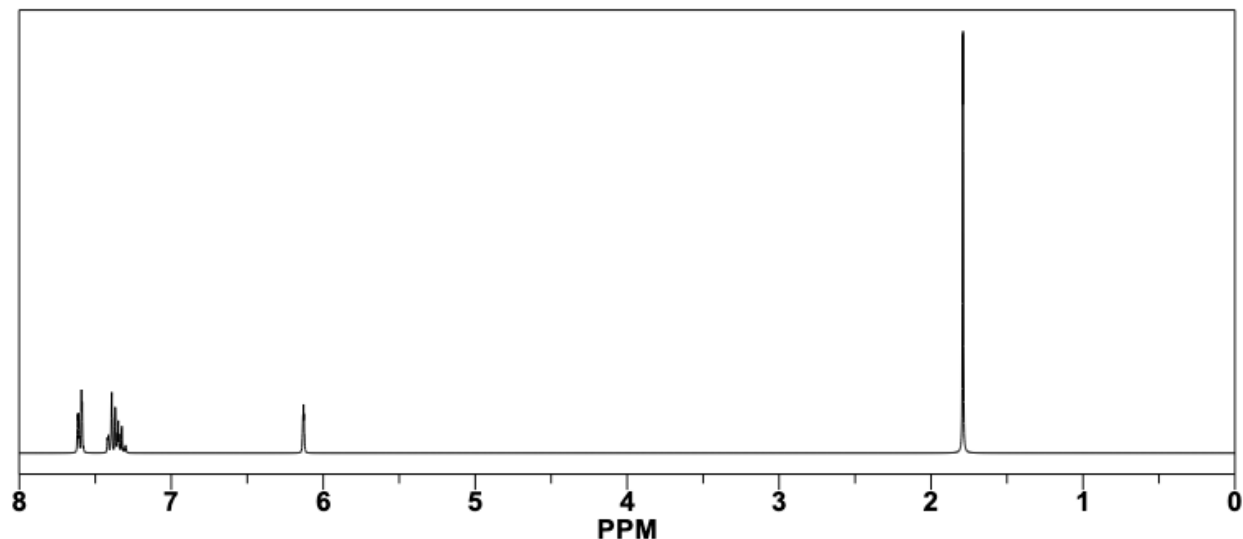
1. For each molecule below, determine if it can undergo E1 to form an alkene when combined with a non-nucleophilic acid. Briefly explain why or why not.



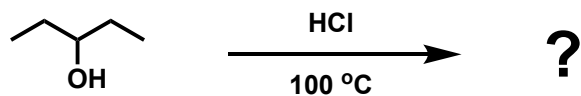
2. A substrate is combined with phosphoric acid and heated to form a single product. An NMR spectrum was then acquired of the product. Which molecule below was the substrate? Assume that the only signals that were observed for this molecule are in the spectrum are shown.



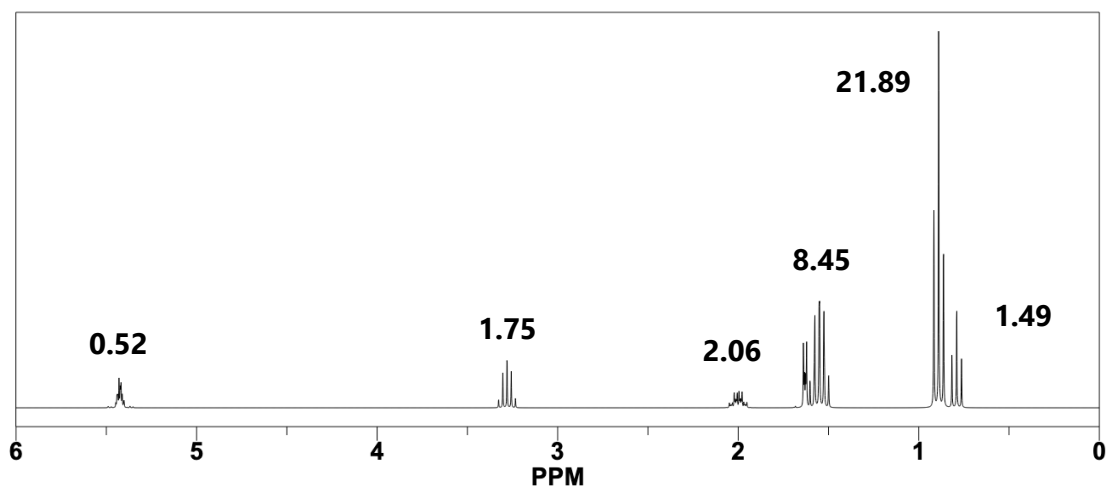
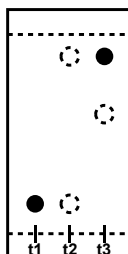
3. A substrate is combined with phosphoric acid and heated to form a single product. An NMR spectrum was then acquired of the product. Which molecule below was the substrate? Assume that the only signals that were observed for this molecule are in the spectrum are shown.



4. Consider the reaction below:



The following TLC was developed at various time points during the reaction (t1, t2, t3) and then an NMR spectrum was taken of the final reaction mixture:



- At what time (t1, t2, or t3) is all the 3-pentanol consumed?
- What product(s) are formed during the reaction?
- What product is formed first?
- Assuming that only two products are formed, what percentage of the final mixture is the major product?